CONICAL FLOW:

Code for Calculating Properties of the Conical Flow Field Past a Cone at Supersonic Speed

APRIL 17, 2025

ARO 3111.02: COMPUTER ASSIGNMENT REPORT

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Executive Summary

Per the instructions provided, my task was to develop a computer program in a high-level computer language, such as MATLAB or C++, to numerically calculate the properties of the conical flow field generated by a right circular cone flying at a supersonic Mach number. The cone is to have a pitch angle of zero and a yaw angle of zero. We are evaluating the behavior of this sharp tip cone through an inviscid, perfect gas.

Our ARO3011 class were provided the second-order, nonlinear ordinary differential equation, the Taylor-Maccoll Equation by Dr. Lin and Dr. Ahmadi. The first step was to convert this second order equation into two first-order differential equations. The relevant definitions, values, and equations were also provided. The required method of solving these was by developing an algorithm that would integrate these equations using the 4^{th}-order (Classical) Runge-Kutta method. The calculations begin by using the provided initial conditions at the conical shock wave. An integrating loop would march in small increments towards the surface of the cone. Using oblique shock relations, the loop would cease as soon as the theta component of velocity equals zero. That is because this would indicate that we have arrived at the surface of the cone. Once we determine where the cone surface is at, we now know what the half-angle of the cone is and can continue with the remaining calculations.

We also consider that the conical flow is fully isentropic, the free stream flow ahead of the shock is isentropic, and the only region of non-isentropic flow is across the shock wave, making it adiabatic. This tells us that the total temperature is constant everywhere, but the total pressure will drop across the shock. Using this knowledge, in tandem with equations mentioned in equations from Anderson's textbook portrayed in Summary of Key Equations, locally planar shock relations help us find the flow properties for our graphs.

Information I obtained included the radial and theta components of velocity, velocity itself, post shock Mach number at varying angles including at the surface of the cone, the corresponding intermediate step values requiring normal Mach number components, and temperature, density, and pressure ratios pre and post shock.

Upon comparing my graphs to the graphs provided in Dr. Lin's slides, I found that the information was identical. Though the graphs provided did not have the y-axis values labeled, the expected range was later provided via email. This allowed for a better comparison, confirming high accuracy.

List of Symbols

Below is a figure of the cone that includes the conical shock wave, the coordinate system reflecting how the change in angle is directed, angles, and velocity components.

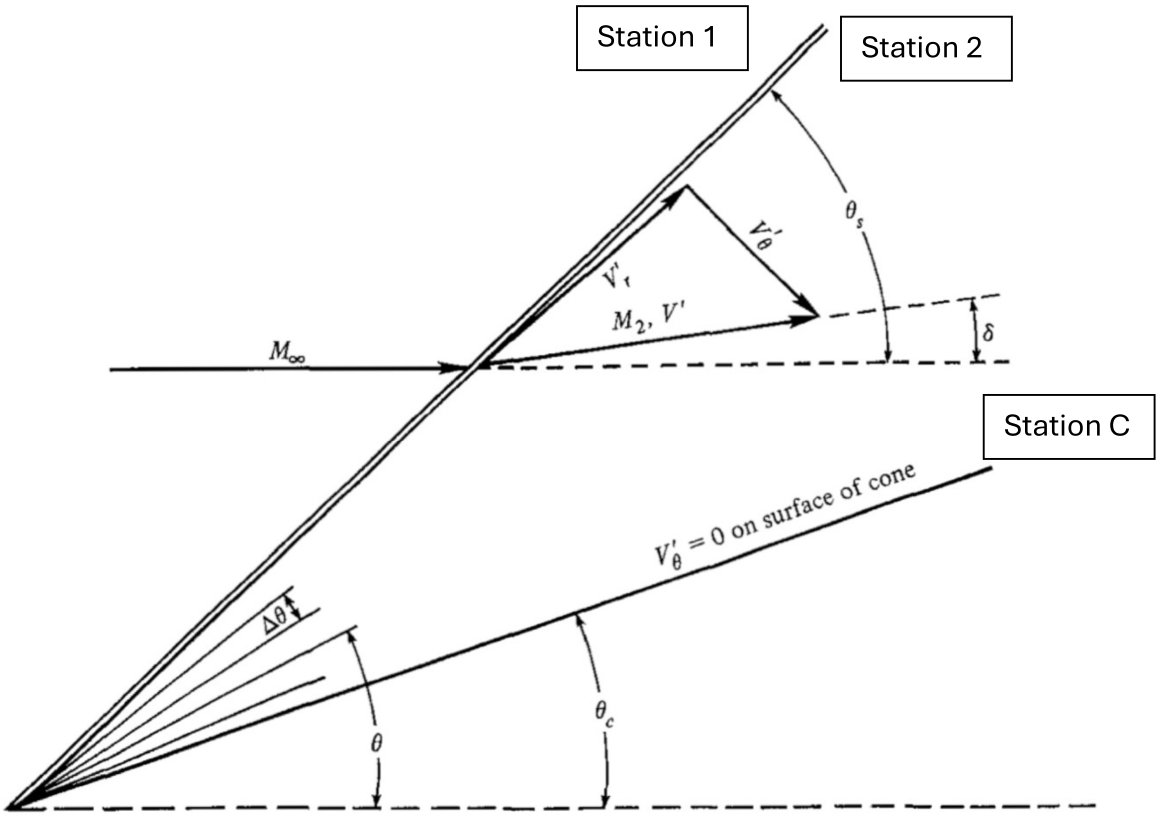

Figure 10.4 | Geometry for the numerical solution of flow over a cone.

<u>List of Symbols:</u>

M_1 = Mach number at station 1 *or* M_∞, Mach number of free stream

M_2 = Mach number at station 2

M_{n1} = normal component of Mach number 1

M_{n2} = normal component of Mach number 2

V' = normalized velocity

V'_r = radial component of normalized velocity

V'_θ = theta-component of normalized velocity

θ_s = angle theta of the shock wave

θc = half-angle of the cone

$\Delta\theta$ = increments of theta marched away from the shock wave and towards the cone surface

δ = deflection of M_2 as it crosses the shock wave

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Summary of Key Equations

Taylor-Maccoll Equations with Initial Conditions:

$$\frac{dV_r}{d\theta} = V_\theta$$

$$\frac{d^2V_r}{d\theta^2} = \frac{V_\theta^2 V_r - \frac{\gamma-1}{2}(1-V_r^2-V_\theta^2)(2V_r+V_\theta\cot\theta)}{\frac{\gamma-1}{2}(1-V_r^2-V_\theta^2)-V_\theta^2}$$

Herein the velocity components are normalized with $V_{\max}$

Integrating procedurres:

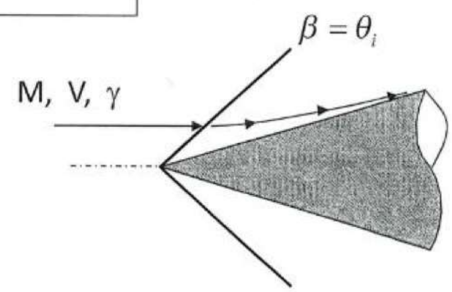
$$u_1 = V_r, \qquad \frac{du_1}{d\theta} = f_1$$

$$u_2 = \frac{dV_r}{d\theta}, \qquad \frac{du_2}{d\theta} = f_2$$

Initial conditions at the shock $\theta_i = \beta$

$$V_r = V_\infty \cos\beta = V_\infty \cos\theta_i$$

$$V_\theta = V_\infty \sin\theta_i \left[\frac{(\gamma-1)M_\infty^2\sin^2\theta_i + 2}{(\gamma+1)M_\infty^2\sin^2\theta_i}\right]$$



• *On the cone surface:* $V_\theta = 0$

Runge-Kutta method to integrate in the 4th order:

Given:

$$u_1' = \frac{du_1}{dt}$$

$$u_2' = \frac{du_2}{dt}$$

$$u_1' = f_1(t, u_1, u_2), \quad u_1(a) = \alpha_1,$$
$$u_2' = f_2(t, u_1, u_2), \quad u_2(a) = \alpha_2,$$
$$a \le t \le b,$$

Find u_1, and u_2

t2=t1+h

$$k_{1,1} = hf_1(t_i, w_{1,i}, w_{2,i}),$$
$$k_{1,2} = hf_2(t_i, w_{1,i}, w_{2,i}),$$

$$k_{2,1} = hf_1\left(t_i + \frac{h}{2}, w_{1,i} + \frac{k_{1,1}}{2}, w_{2,i} + \frac{k_{1,2}}{2}\right),$$
$$k_{2,2} = hf_2\left(t_i + \frac{h}{2}, w_{1,i} + \frac{k_{1,1}}{2}, w_{2,i} + \frac{k_{1,2}}{2}\right),$$

$$k_{3,1} = hf_1\left(t_i + \frac{h}{2}, w_{1,i} + \frac{k_{2,1}}{2}, w_{2,i} + \frac{k_{2,2}}{2}\right),$$
$$k_{3,2} = hf_2\left(t_i + \frac{h}{2}, w_{1,i} + \frac{k_{2,1}}{2}, w_{2,i} + \frac{k_{2,2}}{2}\right),$$

$$k_{4,1} = hf_1(t_i + h, w_{1,i} + k_{3,1}, w_{2,i} + k_{3,2}),$$
$$k_{4,2} = hf_2(t_i + h, w_{1,i} + k_{3,1}, w_{2,i} + k_{3,2}),$$

$$w_{1,i+1} = w_{1,i} + \frac{1}{6}(k_{1,1} + 2k_{2,1} + 2k_{3,1} + k_{4,1}),$$
$$w_{2,i+1} = w_{2,i} + \frac{1}{6}(k_{1,2} + 2k_{2,2} + 2k_{3,2} + k_{4,2}),$$

where $u_1(t_i) \approx w_{1,i}$, $u_2(t_i) \approx w_{2,i}$.

The following are the equations taken directly from Anderson's Modern Compressible Flow text. They show the chapter and equation number. The last equation, 138, was taken from the NACA 1135 report.

$$\boxed{\frac{T_o}{T} = 1 + \frac{\gamma - 1}{2} M^2} \tag{3.28}$$

$$\boxed{\frac{p_o}{p} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\gamma/(\gamma-1)}} \tag{3.30}$$

$$\boxed{\frac{\rho_o}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{1/(\gamma-1)}} \tag{3.31}$$

$$\boxed{M_2^2 = \frac{1 + [(\gamma - 1)/2]M_1^2}{\gamma M_1^2 - (\gamma - 1)/2}} \tag{3.51}$$

$$\boxed{\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2}} \tag{3.53}$$

$$\boxed{\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}\left(M_1^2 - 1\right)} \tag{3.57}$$

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)\left(\frac{\rho_1}{\rho_2}\right) \tag{3.58}$$

$$M_{n_1} = M_1 \sin\beta \tag{4.7}$$

$$M_{n_2}^2 = \frac{M_{n_1}^2 + [2/(\gamma - 1)]}{[2\gamma/(\gamma - 1)]M_{n_1}^2 - 1} \tag{4.10}$$

$$M_2 = \frac{M_{n_2}}{\sin(\beta - \theta)} \tag{4.12}$$

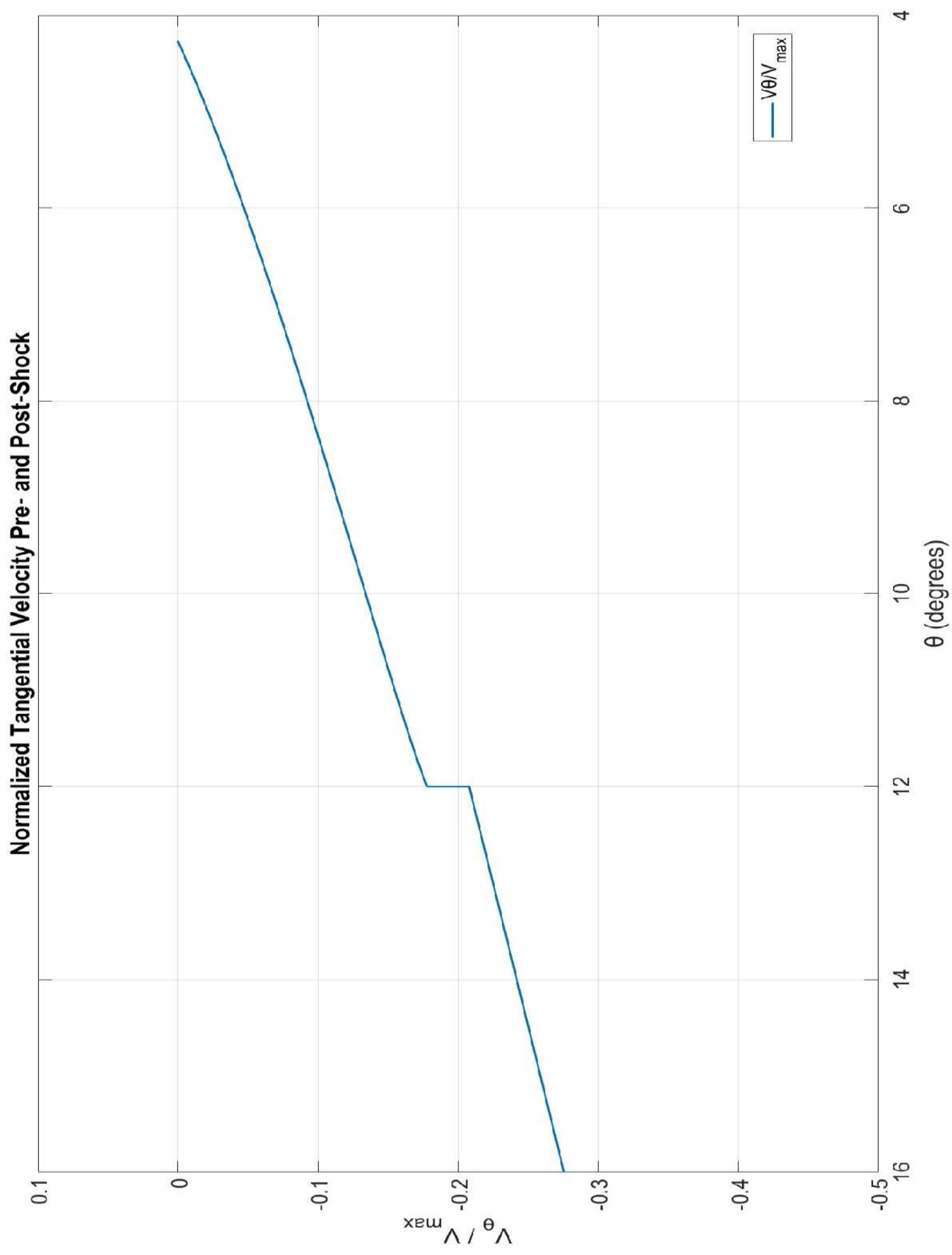
$$\boxed{\tan\theta = 2\cot\beta\left[\frac{M_1^2\sin^2\beta - 1}{M_1^2(\gamma + \cos 2\beta) + 2}\right]} \tag{4.17}$$

$$\frac{V}{V_{\max}} \equiv V' = \left[\frac{2}{(\gamma - 1)M^2} + 1\right]^{-1/2} \tag{10.16}$$

$$\cot\delta = \tan\theta\left[\frac{(\gamma + 1)M_1^2}{2(M_1^2\sin^2\theta - 1)} - 1\right] \tag{138}$$

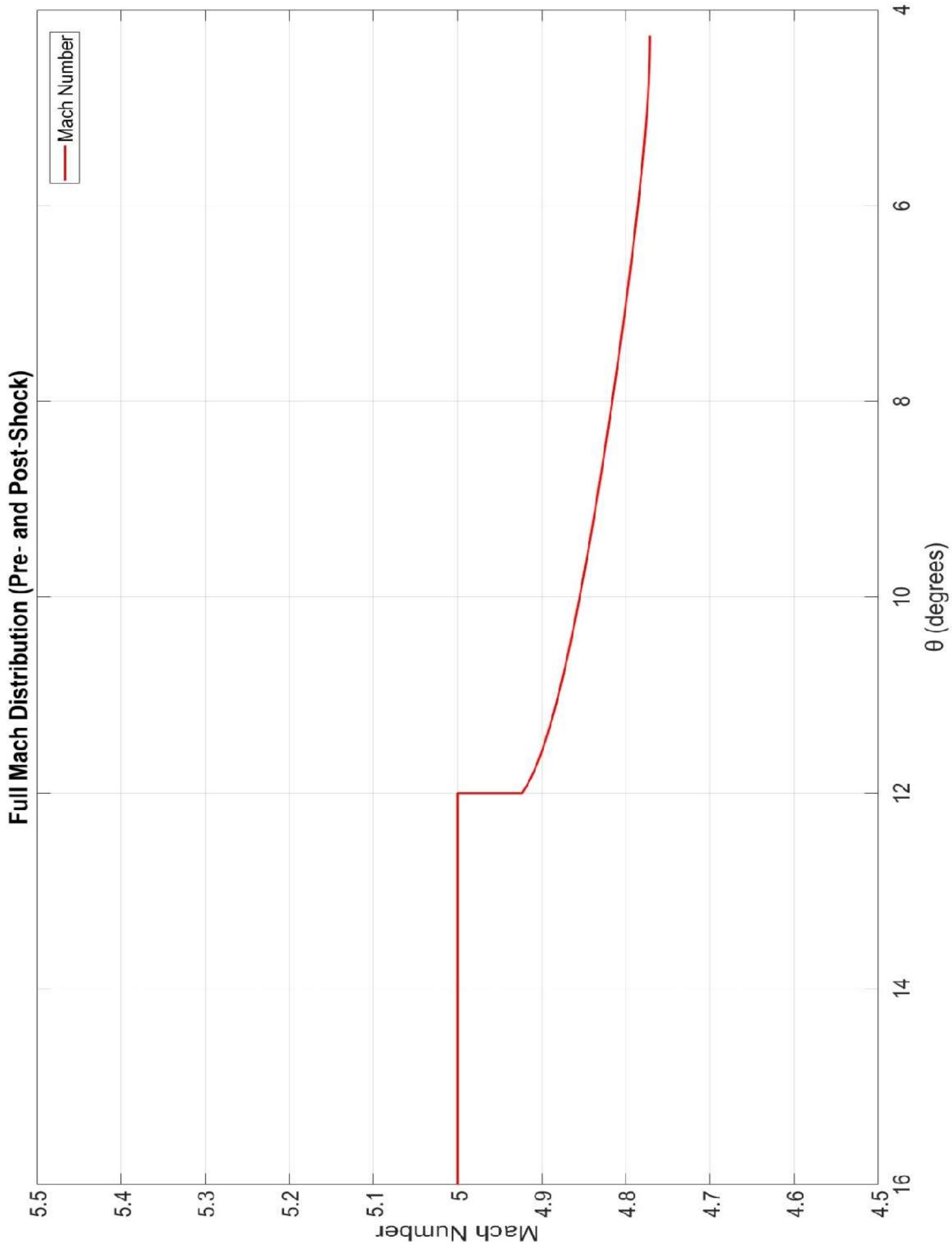
Plot 1: V_θ v. θ

This plot demonstrates the normalized theta component of velocity plotted against the corresponding angle. We see the curve come to a halt at θ = 4.3˚.



Plot 2: M v. θ

This plot demonstrates the Mach values at different values of θ. X-axis limits and Y-axis limits were provided by Dr. Lin and Dr. Ahmadi and are shown in the plot.



Plot 3: p/p1, T/T1, ρ/ρ1 v. θ

The aerodynamic flow property ratios are plotted against θ, where values greater than θ = 12° are equal to 1 because p = p1, so p/p1 = 1, and so forth.

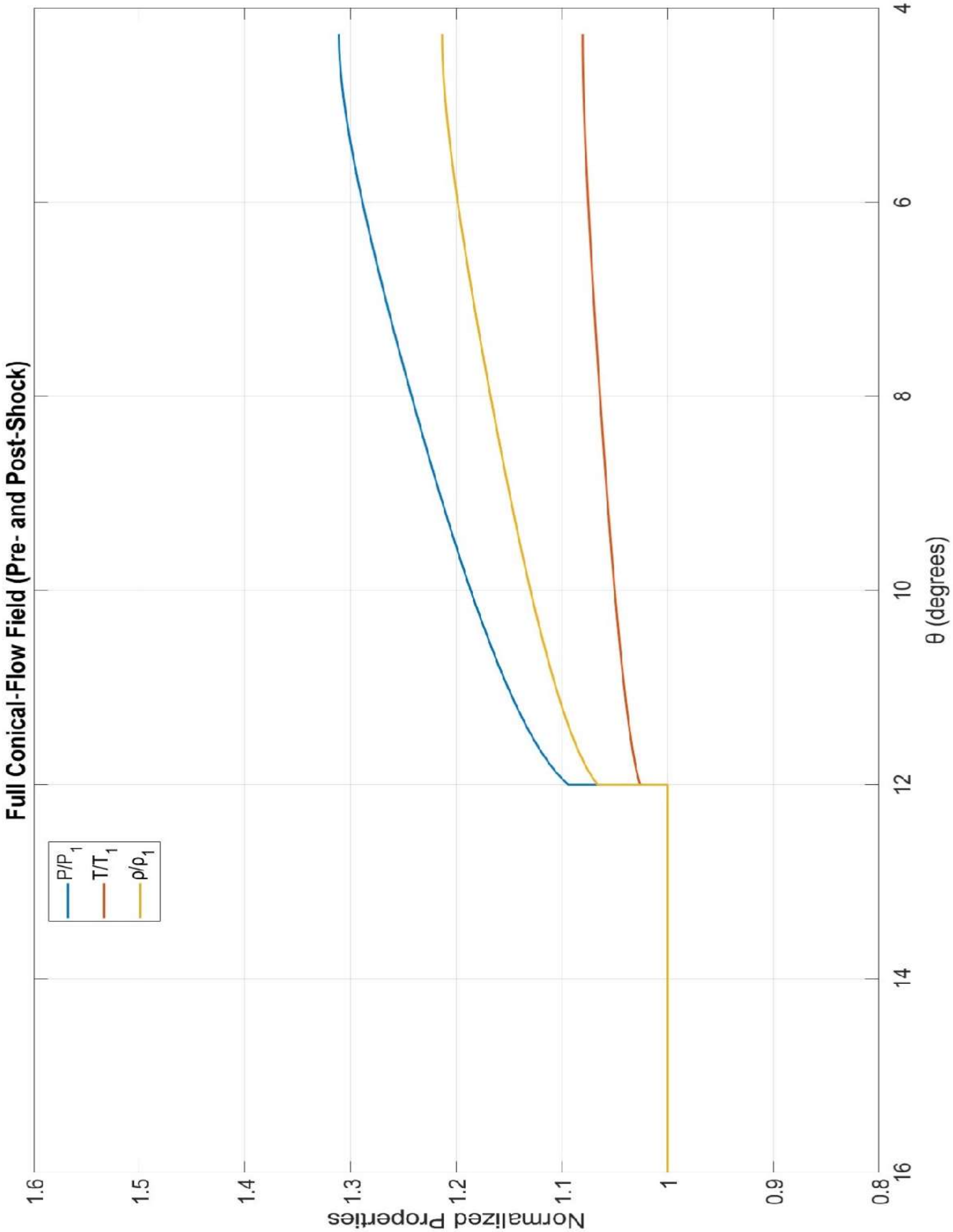


Table of Computed Data

Table of Computed Data (θ from 14° → θ_c, inverted):					
θ (deg)	Mach	P/P1	T/T1	ρ/ρ1	Vθ/Vmax
14	5	1	1	1	-0.2419
13	5	1	1	1	-0.225
12	5	1	1	1	-0.2079
11.9	4.9154	1.1042	1.0288	1.0734	-0.175
11.8	4.9103	1.111	1.0305	1.0781	-0.1727
11.7	4.9064	1.1161	1.0319	1.0816	-0.1708
11.6	4.9003	1.1242	1.034	1.0872	-0.1676
11.5	4.896	1.1298	1.0355	1.0911	-0.1652
11.4	4.8935	1.1332	1.0364	1.0934	-0.1637
11.3	4.8889	1.1395	1.038	1.0977	-0.1608
11.2	4.8861	1.1432	1.039	1.1003	-0.159
11.1	4.8835	1.1467	1.0399	1.1027	-0.1572
11	4.8809	1.1502	1.0408	1.1051	-0.1554
10.9	4.8779	1.1543	1.0419	1.1079	-0.1532
10.8	4.875	1.1583	1.0429	1.1107	-0.151
10.7	4.8722	1.1622	1.0439	1.1133	-0.1489
10.6	4.8689	1.1667	1.0451	1.1164	-0.1462
10.5	4.8657	1.1712	1.0462	1.1194	-0.1436
10.4	4.8657	1.1712	1.0462	1.1194	-0.1436
10.3	4.8626	1.1754	1.0473	1.1224	-0.141
10.2	4.8591	1.1805	1.0486	1.1258	-0.1378
10.1	4.8556	1.1854	1.0498	1.1291	-0.1347
10	4.8556	1.1854	1.0498	1.1291	-0.1347
9.9	4.8522	1.1901	1.051	1.1324	-0.1315
9.8	4.8489	1.1948	1.0522	1.1355	-0.1284
9.7	4.8489	1.1948	1.0522	1.1355	-0.1284
9.6	4.845	1.2003	1.0536	1.1393	-0.1246
9.5	4.845	1.2003	1.0536	1.1393	-0.1246
9.4	4.8413	1.2058	1.0549	1.143	-0.1207
9.3	4.8413	1.2058	1.0549	1.143	-0.1207
9.2	4.8376	1.2111	1.0563	1.1466	-0.1168
9.1	4.8376	1.2111	1.0563	1.1466	-0.1168
9	4.8339	1.2163	1.0576	1.1501	-0.113
8.9	4.8339	1.2163	1.0576	1.1501	-0.113
8.8	4.8304	1.2214	1.0588	1.1536	-0.1091
8.7	4.8304	1.2214	1.0588	1.1536	-0.1091
8.6	4.827	1.2265	1.0601	1.157	-0.1052
8.5	4.8236	1.2314	1.0613	1.1603	-0.1012
8.4	4.8236	1.2314	1.0613	1.1603	-0.1012
8.3	4.8203	1.2363	1.0625	1.1636	-0.0973
8.2	4.8203	1.2363	1.0625	1.1636	-0.0973
8.1	4.817	1.2411	1.0637	1.1668	-0.0933

Table of Computed Data (continued)					
θ (deg)	Mach	P/P1	T/T1	ρ/ρ1	Vθ/Vmax
8	4.817	1.2411	1.0637	1.1668	-0.0933
7.9	4.8138	1.2459	1.0648	1.17	-0.0893
7.8	4.8138	1.2459	1.0648	1.17	-0.0893
7.7	4.8107	1.2505	1.066	1.1731	-0.0853
7.6	4.8107	1.2505	1.066	1.1731	-0.0853
7.5	4.8076	1.2552	1.0671	1.1762	-0.0812
7.4	4.8076	1.2552	1.0671	1.1762	-0.0812
7.3	4.8046	1.2597	1.0682	1.1793	-0.0771
7.2	4.8017	1.2642	1.0693	1.1822	-0.073
7.1	4.8017	1.2642	1.0693	1.1822	-0.073
7	4.7988	1.2685	1.0703	1.1852	-0.0688
6.9	4.7988	1.2685	1.0703	1.1852	-0.0688
6.8	4.796	1.2728	1.0714	1.188	-0.0646
6.7	4.796	1.2728	1.0714	1.188	-0.0646
6.6	4.7932	1.277	1.0724	1.1908	-0.0603
6.5	4.7932	1.277	1.0724	1.1908	-0.0603
6.4	4.7905	1.2811	1.0734	1.1936	-0.0559
6.3	4.7905	1.2811	1.0734	1.1936	-0.0559
6.2	4.788	1.2851	1.0743	1.1962	-0.0515
6.1	4.788	1.2851	1.0743	1.1962	-0.0515
6	4.7855	1.2889	1.0752	1.1987	-0.047
5.9	4.7855	1.2889	1.0752	1.1987	-0.047
5.8	4.7831	1.2926	1.0761	1.2012	-0.0424
5.7	4.7809	1.2961	1.0769	1.2035	-0.0378
5.6	4.7809	1.2961	1.0769	1.2035	-0.0378
5.5	4.7793	1.2986	1.0775	1.2052	-0.0341
5.4	4.7777	1.301	1.0781	1.2067	-0.0304
5.3	4.7777	1.301	1.0781	1.2067	-0.0304
5.2	4.7763	1.3032	1.0786	1.2082	-0.0266
5.1	4.7751	1.3052	1.0791	1.2095	-0.0228
5	4.7751	1.3052	1.0791	1.2095	-0.0228
4.9	4.7739	1.307	1.0795	1.2107	-0.0188
4.8	4.773	1.3085	1.0799	1.2117	-0.0148
4.7	4.7723	1.3095	1.0801	1.2123	-0.0116
4.6	4.7723	1.3095	1.0801	1.2123	-0.0116
4.5	4.7718	1.3102	1.0803	1.2129	-0.0083
4.4	4.7715	1.3108	1.0804	1.2132	-0.0049
4.3	4.7713	1.311	1.0805	1.2134	-0.0015

Appendix A: Hand Calculations

Station 2: Just behind the shock

$M_\infty = 5$, $\gamma = 1.4$, $\theta_s = 12°$

$M_{n_1} = M_1 \cdot \sin\beta$

$= 5 \cdot \sin(12°)$

$= 1.0395585$

$$M_{n_2} = \sqrt{\frac{(1.03956)^2 + (2/(1.4-1))}{(2(1.4)/(1.4-1))(1.03956)^2 - 1}}$$

$= 0.96242368$

$$\delta = \cot^{-1}\left(\tan(12°) \cdot \left[\frac{(1.4+1)5^2}{2(5^2 \cdot \sin^2(12°) - 1)} - 1\right]\right)$$

$= 0.72685636°$

$$M_2 = \frac{0.96242368}{\sin(12 - 0.72685636)}$$

$= 4.9232238$

$$V' = \left[\frac{2}{(1.4-1) \cdot 5^2} + 1\right]^{-1/2}$$

$= 0.91048912$

$V_\theta' = 0.91048912 \cdot \sin(12°)$

$= 0.189301$

$V_r' = 0.890593$

$$\frac{p_2}{p_1} = 1 + \frac{2(1.4)}{1.4+1}(1.0815 - 1) = 1.0952$$

$$\frac{p}{p_{02}} = \left[1 + \frac{1.4-1}{2}(4.92)^2\right]^{-1.4/(1.4-1)} = 0.0020$$

$$\frac{p_2}{p_{02}} = \left[1 + \frac{1.4-1}{2}(0.9619)^2\right]^{-1.4/(1.4-1)} = 0.551977$$

$$\frac{p}{p_1} = \left(\frac{p}{p_{02}} \Big/ \frac{p_2}{p_{02}}\right) \cdot \frac{p_2}{p_1} = 1.0952$$

$$\frac{\rho}{\rho_{02}} = \left(1 + \frac{1.4-1}{2} 4.92^2\right)^{-1/(1.4-1)} = 0.0121$$

$$\frac{\rho_2}{\rho_{02}} = \left(1 + \frac{1.4-1}{2} \cdot 0.9619^2\right)^{-1/(1.4-1)} = 0.6090$$

$$\frac{\rho_2}{\rho_1} = \frac{(1.4+1) \cdot 1.0815}{(1.4-1)(1.0396)^2 + 2} = 1.067$$

$$\frac{\rho}{\rho_1} = \left(\frac{\rho}{\rho_{02}} \Big/ \frac{\rho_2}{\rho_{02}}\right) \cdot \frac{\rho_2}{\rho_1} = 1.068$$

$$\frac{T}{T_{02}} = \left(1 + \frac{1.4-1}{2} 4.92^2\right)^{-1} = 0.171195$$

$$\frac{T_2}{T_{02}} = \left(1 + \frac{1.4-1}{2} 0.9619^2\right)^{-1} = 0.8437$$

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \Big/ \frac{\rho_2}{\rho_1}\right) = \frac{1.0952}{1.067} = 1.0264$$

$$\frac{T}{T_1} = \left(\frac{T}{T_{02}} \Big/ \frac{T_2}{T_{02}}\right) \cdot \frac{T_2}{T_1} = 1.0262$$

Station C: surface of the cone, $\theta = 4.2675°$

$\theta_c = 4.2675°$

$V_r @ \theta_c = 0.8929$

$V^2 = 0.79727$

$$M = \sqrt{\frac{2}{1.4-1}\left(\frac{1}{0.79727} - 1\right)} = 1.126$$

$M_{n_1} = 5 \cdot \sin(12) = 1.0395$

$M_2 = 0.9619$

$$\frac{p_2}{p_1} = 1 + \frac{2.8}{2.4}(1.0815 - 1) = 1.0952$$

$$\frac{p}{p_{02}} = \left(1 + \frac{1.4-1}{2}(1.126)^2\right)^{-1.4/(1.4-1)} = 0.4609$$

$$\frac{p_2}{p_{02}} = \left(1 + \frac{1.4-1}{2}(0.9619)^2\right)^{-3.5} = 0.5135$$

$$\frac{p}{p_1} = \frac{0.4609}{0.5135} \cdot 1.0952 = 0.983$$

$$\frac{\rho}{\rho_{02}} = \left(1 + \frac{1.4-1}{2}(1.126)^2\right)^{-1/(1.4-1)} = 0.5779$$

$$\frac{\rho_2}{\rho_{02}} = \left(1 + \frac{1.4-1}{2}(0.9619)^2\right)^{-2.5} = 0.6090$$

$$\frac{\rho_2}{\rho_1} = \frac{2.4(1.0815)}{0.4 \times 1.0815 + 2} = 1.067$$

$$\frac{\rho}{\rho_1} = \frac{0.5779}{0.6090} \cdot 1.067 = 1.012$$

$$\frac{T}{T_{02}} = \left(1 + \frac{1.4-1}{2}(1.126)^2\right)^{-1} = 0.7975$$

$$\frac{T_2}{T_{02}} = \left(1 + \frac{1.4-1}{2}(0.9619)^2\right)^{-1} = 0.8437$$

$$\frac{T_2}{T_1} = \frac{1.0952}{1.067} = 1.0264$$

$$\frac{T}{T_1} = \frac{0.7975}{0.8437} \cdot 1.0264 = 0.971$$

θ	Mach	P/P1	T/T1	ρ/ρ1	Vθ/Vmax
11.9	4.9154	1.1042	1.0288	1.0734	-0.175
4.3	4.7713	1.331	1.0805	1.2134	-0.0015

The difference in values here is likely due to human error when rounding in between steps, whereas the MATLAB code would have superior accuracy.

Appendix B: Program Listing

ARO3111ComputerAssignment.m

```
clear all
clc
% This code will numerically calculate the properties of the conical flow
% field generated by a right circular cone flying at supersonic Mach number
% at zero pitch and zero yaw through an inviscid and perfect gas.

theta_s_d=12; %Conical shock wave angle in degrees
theta_s_r = theta_s_d*pi/180; %Conical shock wave angle, converted from deg to rad
M1 = 5; % Free stream Mach number
gamma = 1.4; % Ratio of specific heats

%The half-angle of the cone is calculated from ConeAngleFinder file
[theta_c_r] = ConeAngleFinder(0,theta_s_r,M1,gamma,'rad');

%Using oblique shock relations to get M behind shock
Mn1 = M1*sin(theta_s_r); % Normal Mach number pre-shock
Mn2 = sqrt((1 + ((gamma-1)/2)*Mn1^2)/((gamma*Mn1^2) - ((gamma-1)/2))); % Normal Mach
number post-shock
M2 = Mn2/(sin(theta_s_r-theta_c_r)); % Post-shock Mach number

%Initial conditions
V = ((2/((gamma-1)*M2^2))+1)^(-1/2);
Vtheta = -V*(sin(theta_s_r-theta_c_r));
Vr = V*(cos(theta_s_r-theta_c_r));

%Range that theta will span, along with initial conditions of Vr and Vr'
thetarange = [theta_s_r; 1e-10]; % Will integrate from shock angle to about 0 degrees
V_i = [Vr; Vtheta]; % Initial
%Solutions from Runge-Kutta Method, using defined function
[solution1, solution2] = RungeKutta4(Vr, Vtheta, theta_c_r, theta_s_r, gamma);
options = odeset('Events',@EVENTS, 'Reltol', 2.22045e-14); %limiting parameters
solution = ode23t(@TM,thetarange,V_i,options,gamma);

%Calculate angle and Mach number at the cone
[n,m] = size(solution.y); %Retrieves the size of the solution array
if (n > 0 && m > 0 && isempty(solution.ie) ~= 1) %If the value converges, the
following are final values:
        theta_c_r = solution.xe;
        theta_c_d = theta_c_r.*(180/pi); %final cone angle [deg]
        Vc2 = solution.ye(1)^2 + solution.ye(2)^2;
        Mc = ((1.0/Vc2-1)*(gamma-1)/2)^-0.5; %Mach number at cone surface
else
        fprintf('Error 404: Cone angle not found!\n');
end

solution_V2 = (solution.y(1,:).^2 + solution.y(2,:).^2)'; %Total velocity squared
solution_Mc = ((1./solution_V2-1).*(gamma-1)/2).^(-0.5); %Mach number at the surface
of the cone
solution_X_rad = solution.x'; %Integration angles in radians
solution_X_deg = (solution.x').*(180/pi); %Integration angles converted to degrees
```

```
%Results
fprintf('When given:\nM_freestream = %0.3f, gamma = %0.3f, theta_shock = %0.3f
deg\n\nFinal Mach numbers and half-angle of cone are::\n', M1, gamma,
180*theta_s_r/pi);
fprintf('M2 = %0.3f\n', M2);
fprintf('M_cone = %0.3f\n', Mc);
fprintf('theta_c = %0.3f degrees\n', theta_c_d);

%Formulas for aerodynamic property calculations:

%Oblique shock relations, uses Mach 1 normal component
rho2_rho1 = (gamma+1)*Mn1^2/((gamma-1)*Mn1^2+2);
P2_P1 = 1 + 2*gamma*(Mn1^2-1)/(gamma+1);
T2_T1 = P2_P1/rho2_rho1;

%Uses Mach number at surface of cone
T02_T = 1+(gamma-1)*solution_Mc.^2/2;
P02_P = (1+(gamma-1)*solution_Mc.^2/2).^(gamma/(gamma-1));
rho02_rho = (1+(gamma-1)*solution_Mc.^2/2).^(1/(gamma-1));

%Uses Mach 2
T02_T2 = 1+(gamma-1)*M2^2/2;
P02_P2 = (1+(gamma-1)*M2^2/2)^(gamma/(gamma-1));
rho02_rho2 = (1+(gamma-1)*M2^2/2)^(1/(gamma-1));

%Required aerodynamic properties for charts:
P_P1 = P02_P2*P2_P1./P02_P;
T_T1 = T02_T2*T2_T1./T02_T;
rho_rho1 = rho02_rho2*rho2_rho1./rho02_rho;

%Define pre-shock region
theta_start_deg = 20;
delta_theta = -0.1;
theta_pre = theta_start_deg : delta_theta : theta_s_d; % e.g. 20 → 12 deg
M_pre = M1 * ones(size(theta_pre));
Vtheta_pre = -sind(theta_pre);

%Freestream (pre-shock) ratios
P_pre = ones(size(theta_pre));
T_pre = ones(size(theta_pre));
rho_pre = ones(size(theta_pre));

%Gather post-shock solution
theta_post = solution_X_deg; % e.g. 12 → 4.267 deg
P_post = P_P1;
T_post = T_T1;
rho_post = rho_rho1;
M_post = solution_Mc;
Vtheta_post = solution.y(2,:);

%Concatenate into full fields
theta_full = [theta_pre, theta_post.'];
P_full = [P_pre, P_post.'];
T_full = [T_pre, T_post.'];
```

```
rho_full = [rho_pre, rho_post.'];
M_full = [M_pre, M_post.'];
Vtheta_full = [Vtheta_pre, Vtheta_post];

figure(1);
plot(theta_full, Vtheta_full, 'LineWidth',1.5, 'DisplayName','Vθ/V_{max}');
set(gca,'xdir','reverse');
xlim([4, 16]);
ylim([-0.5 0.1]);
xticks([4:2:16]);
grid on;
xlabel('θ (degrees)');
ylabel('V_θ / V_{max}');
title('Normalized Tangential Velocity Pre- and Post-Shock');
legend('Location','best');

figure(2);
plot(theta_full, M_full, 'r', 'LineWidth', 1.5, 'DisplayName', 'Mach Number');
set(gca,'xdir','reverse');
xlabel('θ (degrees)');
ylabel('Mach Number');
legend('Location','best');
title('Full Mach Distribution (Pre- and Post-Shock)');
xlim([4 16]);
ylim([4.5 5.5]);
grid on
xticks([4:2:16]);

figure(3);
plot(theta_full, P_full, 'LineWidth', 1.5, 'DisplayName','P/P_1'); hold on
plot(theta_full, T_full, 'LineWidth', 1.5, 'DisplayName','T/T_1');
plot(theta_full, rho_full, 'LineWidth', 1.5, 'DisplayName','ρ/ρ_1');
set(gca,'xdir','reverse');
xlabel('θ (degrees)');
ylabel('Normalized Properties');
legend('Location','best');
title('Full Conical-Flow Field (Pre- and Post-Shock)');
xlim([4 16]);
ylim([0.8 1.6])
grid on
xticks([4:2:16]);
```

ConeAngleFinder.m

```
function [solution] = ConeAngleFinder(thetaInf,betaInf,MInf,gamma,degrad)
%Obtain the flow deflection angle from the shock wave angle and freestream
%Mach number where:
%thetaInf = cone angle to horizontal
%betaInf = Shockwave angle to horizontal
%MInf = freestream Mach number
%gamma = Ratio of specific heats
%degrad = specify whether calculations are in degrees or radians
%solution = solution
if (strcmp(degrad,'deg'))
```

```
    thetaInf = thetaInf*(pi/180);
    betaInf  = betaInf*(pi/180);
elseif (strcmp(degrad,'rad'))
end
%solve for theta:
if (thetaInf == 0)
    if (betaInf <= asin(1/MInf))
        solution = 0;
        return;
    end
    %Simplifying fractions
    A = ((MInf*sin(betaInf))^2)-1;
    B = (MInf^2*(gamma+cos(2*betaInf)))+2;

%wedge angle from the theta-beta-Mach relation
theta = atan(2*cot(betaInf)*(A/B));
    if (strcmp(degrad,'deg'))
        solution = theta*(180/pi);
    elseif (strcmp(degrad,'rad'))
        solution = theta;
    end
end
```

RungeKutta4.m

```
function [u1, u2] = RungeKutta4(V1, V2, Theta_c, Theta_s, gamma)
%V1 = radial velocity
%V2 = angular velocity
%Theta_c = cone angle
%Theta_s = wave angle

h = -0.1;
x = Theta_s:h:Theta_c;
A = (gamma-1)/2;

%Initialize vectors
u1 = zeros(1,length(x));
u2 = zeros(1,length(x));

%Initial conditions for ODEs
    u1(1) = V1;
    u2(1) = V2;
    t(1)  = 0;

%Define functions as per Dr.Lin and Dr.Ahmadi's notes
    F1 = @(t, u1, u2) u2;
    F2 = @(t, u1, u2) (u1*u2^2-A*(1-u1^2-u2^2)*(2*u1+u2*cot(Theta_s)))/(A*(1-u1^2-
u2^2) - u2^2);

%Runge-Kutta method loop

for i=1:(length(x)-1)

    k1 = h*F1(t, u1(i), u2(i));
```

```
    m1 = h*F1(t, u1(i), u2(i));

    k2 = h*F2(t+0.5*h, u1(i)+0.5*h, u2(i)+0.5*h*k1);
    m2 = h*F2(t+0.5*h, u1(i)+0.5*h, u2(i)+0.5*h*k1);

    k3 = h*F1(t+0.5*h, (u1(i)+0.5*h), (u2(i)+0.5*h*k2));
    m3 = h*F2(t+0.5*h, (u1(i)+0.5*h), (u2(i)+0.5*h*k2));

    k4 = h*F1(t+h, (u1(i)+h),(u2(i)+k3*h));
    m4 = h*F2(t+h, (u1(i)+h),(u2(i)+k3*h));

    % main equation
    u1(i+1) = u1(i) + (1/6)*(k1+2*k2+2*k3+k4)*h;
    u2(i+1) = u2(i) + (1/6)*(m1+2*m2+2*m3+m4)*h;

end
end
```

Events.m

```
function [val,last,dir] = EVENTS(theta,z,gam)
%This sets the parameters necessary to stop the ODE solution when the cone wall is
reached
%theta=integration angle [rad]
%z=angular and radial velocities
%gam=ratio of specific heats
%val=value of the ith event function
%last=integration terminates at a zero of the event function if this value is 1,
otherwise it's 0
%dir=if all zeros are to be located, 0, if zeros where event fnc decreasing, -1, if
zeros where event fnc increasing, +1

val= 1; %no event initiated
last= 1; %stops after event 1 occurs
dir= 0; %get zeros from inc/dec

%To initiate when angular velocity crosses axis, from pos to neg
%angular velocity is zero at the cone surface

if (z(2) > 0)
    val = 0; %Should angular velocity z(2) cross from being negative to positive,
initiate event
end
```

TM.m

```
function [z0] = TM(theta,z,gamma)
%State space representation of Taylor Maccoll equations where:
%theta = integration angle
%z = angular and radial velocities
%gamma = ratio of specific heats
%z0 = solution array
```

```
%Initialize solution vector
z0 = zeros(2,1);
A = (gamma-1)/2;
num = (-2*A*z(1)) - (A*z(2)*cot(theta)) + (2*A*z(1)^3) + (A*z(1)^2*z(2)*cot(theta)) +
(2*A*z(1)*z(2)^2) + (A*z(2)^3*cot(theta)) + (z(1)*z(2)^2);
den = A*(1-z(1)^2-z(2)^2) - z(2)^2;

% State-space representation
z0(1) = z(2);
z0(2) = num/den;
```